Taylor Polynomial: for nth degree polynomial

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

$$P_0(x) = f(x_0)$$

$$P_1(x) = f(x_0) + f'(x_0)(x-x_0)$$

$$P_2(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2$$

Taylor Series: If f has derivatives of all order at x_0, then we call the Taylor Series

$$f(x_0) + f'(x_0)(x-x_0) + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \cdots$$

Ex: Find Taylor Series of the function $f(x) = \cos x$

at $x_0 = \frac{\pi}{2}$ and find $\boxed{P_0, P_1, P_2, P_3, P_4}$ polynomials.

$$f\left(\frac{\pi}{2}\right) = 0$$

$$f'(x) = -\sin x, \quad f'\left(\frac{\pi}{2}\right) = -1$$

$$f''(x) = -\cos x \quad f''\left(\frac{\pi}{2}\right) = 0$$

$$f'''(x) = \sin x, \quad f'''\left(\frac{\pi}{2}\right) = 1$$

$$f^{(IV)}(x) = \cos x, \quad f^{(IV)}\left(\frac{\pi}{2}\right) = 0.$$

$$f(x) \approx 0 + (-1)\left(x - \frac{\pi}{2}\right) + \frac{1}{3!}\left(x - \frac{\pi}{2}\right)^3 + \cdots$$

$$P_0 = f\left(\frac{\pi}{2}\right) = 0$$

$$P_1 = f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right)$$

$$= -\left(x - \frac{\pi}{2}\right)$$

$$P_2 = f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right)\left(x - \frac{\pi}{2}\right) + \frac{f''\left(\frac{\pi}{2}\right)}{2!}\left(x - \frac{\pi}{2}\right)^2$$

$$= -\left(x - \frac{\pi}{2}\right) + 0$$

P_3

P_4 H.W.

H.W. $f(x) = 2x^3 + 7x^2 - x - 1$

Expand the function $f(x)$ in power of $(x+2)$ in infinite Taylor series.

x_0

$x+2$

-2

$f(x) = e^{ax} \longrightarrow (x-1)$

$f(x) =$

H.W. 1. $\cos x$, $\sin x$ at $x = \frac{\pi}{2}$

2. $f(x) = \frac{1}{x+2}$ at $x = 4$

3. Practice sheet.

4. Anton Chapter 9.7 => (17-24)

Chapter 9.8 => (11-18).

Maclaurin Series: $x_0 = 0$

$$f(x) \approx f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

Maclaurin Polynomial:

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

Find Maclaurin polynomial P_0, P_1, P_2, P_3 of

$f(x) = e^x \cos x$

$f(x) = \frac{1}{1-2x}$

H.W.

1. $y = \tan x$, Expand y in the power of x → Maclaurin Series.

2. Expand the function $f(x) = \ln\left(\frac{1+x}{1-x}\right)$ in infinite Maclaurin Series.

3. $f(x) = e^{\cos x}$

4. Practice

5. 9.7 (7-12)
9.8 (1-6) → Anton

Techniques of Differentiation:

1. $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

$n(n-1)(n-2)(n-3)\ldots 3\cdot 2\cdot 1 = n!$

2. $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

$y = (ax+b)^n$, nth differentiation

$y_1 = n(ax+b)^{n-1}\cdot a = an(ax+b)^{n-1}$.

$y_2 = an(n-1)(ax+b)^{n-1-1}\cdot a = a^2 n(n-1)(ax+b)^{n-2}$

$\vdots$

$y_n = a^n n!(ax+b)^{n-n} = a^n n!$

1. $y = x^n$
2. $y = \frac{1}{x+a}$
3. $y = e^{ax}$
4. $y = \sin(ax+b)$
5. $y = \cos(ax+b)$

Must try.

Implicit function:

$$x^2 + y^2 = 5$$

$$\Rightarrow 2x + 2y\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

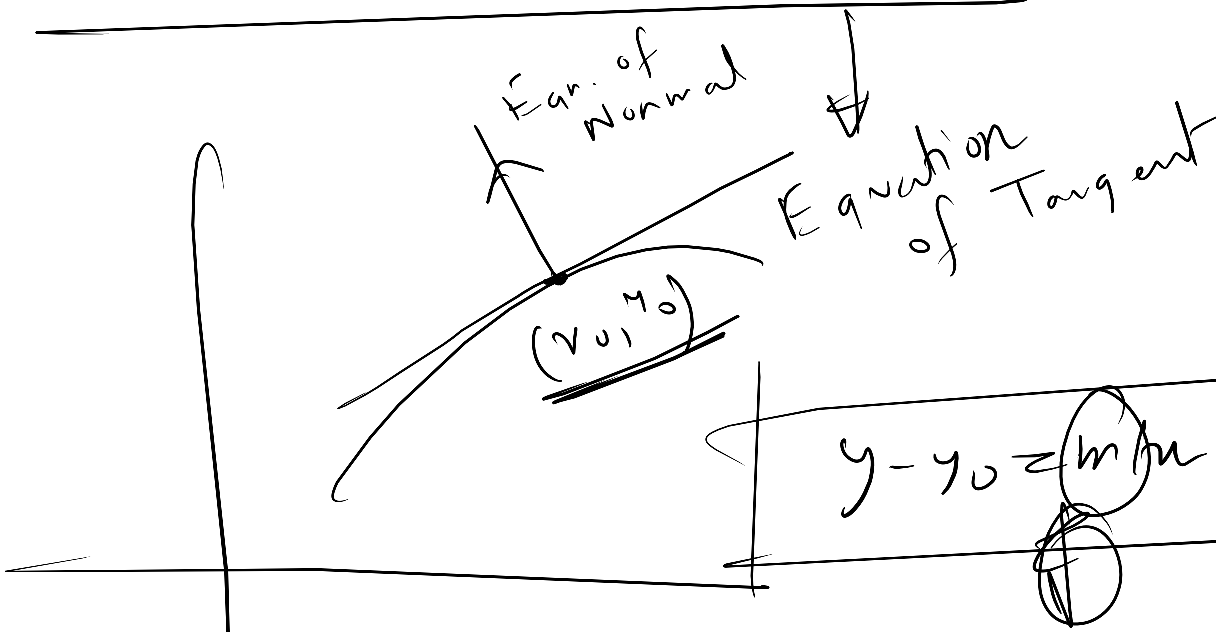
Explicit function

$$y = f(x)$$

$$y = x^2$$

$$\frac{dy}{dx} = 2x$$

Equation of Tangent:



$$\boxed{y - y_0 = m(x - x_0)}$$

$$y - y_0 = \left.\frac{dy}{dx}\right|_{(x_0, y_0)} (x - x_0)$$

Equation of Normal

$$y - y_0 = \frac{-1}{\left.\frac{dy}{dx}\right|_{(x_0, y_0)}} (x - x_0)$$

Ex-05 Find Eqn of tangent & Normal at the point $(1, 2)$ on curve $3x^4 - x^2y + 2y^3 = 0$ ...... (1)

$x_0 \; y_0$

$$12x^3 - \left[x^2 \frac{dy}{dx} + 2xy\right] + 6y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow (6y^2 - x^2)\frac{dy}{dx} = +2xy - 12x^3$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x(y - 6x^2)}{6y^2 - x^2}$$

$$\left.\frac{dy}{dx}\right|_{(1,2)} = \frac{2(2-6)}{6 \cdot 4 - 1} = -\frac{8}{23}$$

Equation of tangent:

$$y-2 = \boxed{-\frac{8}{23}}(x-1).$$

Equation of Normal:

$$y-2 = +\frac{23}{8}(x-1)$$

Logarithmic Differentiation:

$$\ln x^a = a\ln x$$

$$\underbrace{x^y = y^x}$$

$$\Rightarrow \ln x^y = \ln y^x$$

$$\Rightarrow \underset{u}{y}\,\underset{v}{\ln x} = \underset{u}{x}\,\underset{v}{\ln y}$$

$$\Rightarrow \ln x\frac{dy}{dx} + \frac{y}{x} = \ln y + x\frac{1}{y}\frac{dy}{dx}$$

$$\Rightarrow \left(\ln x - \frac{x}{y}\right)\frac{dy}{dx} = \ln y - \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\ln y - y/x}{\ln x - \frac{x}{y}}$$

H.W.

(i) $\cos(x+y) = e^{x+y}$

(ii) $xy + e^{y} = 0$

$\frac{dy}{dx}$

(iii) $(\sin x)^{\cos x} + (\cos x)^{\sin x}$

$\frac{dy}{dx}$

y_1 y_2

$y_1 = (\sin x)^{\cos x}$ $y_2 = (\cos x)^{\sin x}$

$\ln(xy) = \ln x + \ln y$ $\ln x^{a} = a\ln x$

$\ln\left(\frac{x}{y}\right) = \ln x - \ln y$

* let $y = (\sin x)^{\cos x}$

$\Rightarrow \ln y = \cos x \ln(\sin x)$

$\Rightarrow \frac{1}{y}\frac{dy}{dx} = -\sin x \ln(\sin x) + \cos x \frac{\cos x}{\sin x}$

$\Rightarrow \frac{dy}{dx} = y\left[\cot x \cos x - \sin x \ln(\sin x)\right]$

$= (\sin x)^{\cos x}\left[\cot x \cos x - \sin x \ln(\sin x)\right].$